

Quantitative EEG Parameters for Prediction of Outcome in Severe Traumatic Brain Injury: Development Study

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Abstract

Monitoring of quantitative EEG (QEEG) parameters in the intensive care unit (ICU) can aid in the treatment of traumatic brain injury (TBI) patients by complementing visual EEG review done by an expert. We performed an explorative study investigating the prognostic value of 59 QEEG parameters in predicting the outcome of patients with severe TBI. Continuous EEG recordings were done on 28 patients with severe TBI in the ICU of Turku University Hospital. We computed a set of QEEG parameters for each patient, and correlated these to patient outcome, measured by dichotomized Glasgow Outcome Scale (GOS) at a follow-up visit between 6 and 12 months, using area under receiver operating characteristic curve (AUC) as a nonlinear correlation measure. For 17 of the 59 QEEG parameters (28.8%), the AUC differed significantly from 0.5, most of these parameters measured EEG power or variability. The best QEEG parameters for outcome prediction were alpha power (AUC = 0.87, $P < .01$) and variability of the relative fast theta power (AUC = 0.84, $P < .01$). The results of this study indicate that QEEG parameters provide useful information for predicting outcome in severe TBI. Novel QEEG parameters with potential in outcome prediction were found, the prognostic value of these parameters should be confirmed in later studies. The results also provide further evidence of the usefulness of parameters studied in preexisting studies.

Keywords

traumatic brain injury, continuous electroencephalography, quantitative electroencephalography, prognosis, outcome

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Introduction

Severe traumatic brain injury (TBI), defined as having an estimated Glasgow Coma Scale (GCS) 3 to 8 at admission, is commonly treated in the intensive care unit (ICU). Multimodal neuromonitoring is needed for optimal and personalized treatment; continuous EEG is often part of the monitoring and is used mainly to detect epileptic seizures and to help prognostication.^{1,2} Prognostic tools are valuable for clinical work; they may guide clinical treatment decisions and allocation of resources.

The gold standard for EEG interpretation is visual review of the original EEG signals by a certified neurophysiologist and quantitative EEG (QEEG) trends are suggested to support this visual analysis.³ Monitoring the cerebral state of critical care patient, especially if nonconvulsive seizures are suspected, requires continuous EEG monitoring lasting for several days.^{2,4} As long-term nonstop visual EEG review by neurophysiologist is not possible in practice, continuous monitoring of QEEG parameters at the bedside can aid in monitoring by alerting the ICU staff and by serving as a trigger for neurologic and radiologic consultations.

The EEG has been used in traumatic coma since the 1940s and first QEEG studies date back to the 1970s.^{5,6} Multiple

QEEG parameters and EEG characteristics have been associated with outcome. For example, monotonous compressed spectral array (CSA),⁷ lack of alpha variability,^{8,9} decreased fractional power in alpha and beta bands,¹⁰ and decrease or no change in alpha and theta power¹¹ have been found to indicate poor outcome. High bispectral index (BIS) values have been associated with good outcome.¹² In a regression analysis, EEG phase, coherence, and amplitude asymmetry have been found to be good predictors of outcome.¹³ Somatosensory evoked potentials (SSEPs) have independently more prognostic power than EEG monitoring, but combining EEG and SSEP monitoring can provide more reliable prognosis than either of the methods individually.^{14,15}

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This explorative study complements these earlier studies by studying a wide range of fully automatically extracted QEEG parameters. The aim was to find QEEG parameters that have value in predicting the patient outcome in severe TBI and are directly applicable to routine online bedside monitoring.

Material and Methods

Patients and Clinical Protocol

All patients with severe TBI, arriving at the ICU of Turku University Hospital during the study period January 2012 to November 2013, were recruited to the study unless they fulfilled the exclusion criteria. The exclusion criteria for the study were: age less than 18 years at study entry; severely injured scalp, preventing the application of monitoring electrodes; patient gained consciousness before the monitoring could be started or patient was expected to gain consciousness rapidly based on clinical improvement; perforating or penetrating mechanism of TBI or blast-induced TBI; inability to live independently due to preinjury brain disease or other medical cause before the injury. Informed consent was obtained from a proxy and the protocol was approved by the Ethical Committee of the Hospital District of Southwest Finland.

TBI was defined as severe based on the admission GCS and the severity was also reflected in the clinical need for treatment at the ICU. The severity of TBI was also assessed based on imaging, scoring the initial computed tomography findings using the Marshall scale.¹⁶ The outcomes of the patients were assessed using the Glasgow Outcome Scale (GOS) evaluated between 6 and 12 months from the injury at a follow-up visit. The outcome was dichotomized to favorable (GOS 4 and 5) and unfavorable (GOS 1-3).

Treatment Protocol

Patients with severe TBI were treated according to the third edition of Brain Trauma Foundation guidelines¹⁷ with local modifications. Cerebral perfusion pressure was targeted above 60 mm Hg, intracranial pressure below 20 mm Hg, and brain tissue oxygen partial pressure above 20 mm Hg. All patients were sedated either with continuous infusion of propofol or midazolam. Midazolam was chosen in cases of hemodynamic instability. High intracranial pressure, refractory to medical and osmotherapy, was treated with ventriculostomy, mild hypothermia (35°C), and/or decompressive hemicraniectomy. Prophylactic antiepileptic medication, fosphenytoin or levetiracetam, was used if the patient suffered from a temporal lobe contusion or hemorrhage.

EEG Recordings

Altogether, EEG was recorded from 31 patients. Three patients did not arrive to the 6-month follow-up visit and were excluded from the study. EEG monitoring was continued for 72 hours unless patient regained consciousness or the treatment at ICU

was terminated for a medical reason. Interruptions in monitoring were accepted due to clinical reasons necessitating a break in the recording. In case a patient was considered unsalvageable and active medical treatments were withdrawn, EEG recording was stopped. If the patient's unconsciousness was prolonged, EEG monitoring could be continued beyond 72 hours if clinically needed.

The cEEG was recorded using NicoletOne monitoring device (CareFusion Nicolet, San Diego, CA, USA). The international 10-20 electrode placement system was used with Ag-AgCl subcutaneous wire electrodes (Ives EEG Solutions, Manotick, Ontario, Canada) and EEG technician inspected electrode contacts daily. For some of the patients attaching all the electrodes was not possible due to the nature of their injury. For these patients either a montage consisting only of anterior electrodes (3 patients), or only of electrodes on one hemisphere (4 patients) was used.

The median duration of EEG monitoring was 65.9 hours. For most of the patients the EEG monitoring lasted at least 60 hours and after 60 hours there is a steep decline in the number of patients that have EEG monitoring still ongoing. We restricted the amount of EEG analyzed from each patient to a maximum of 60 hours (2.5 days), since large differences in the length of the EEG used for computing the QEEG parameters could have affected the results. Majority of the EEG used in this study was recorded 2 to 5 days after the injury. An alternative way to limit the EEG used in the analysis would have been to use only EEG from a certain time period after the injury, for example, 2 to 5 days after the injury, but this would have meant leaving out 5 subjects from an already limited study population. The temporal distribution of all EEG available and the EEG used in the statistical analysis is shown in Figure 1.

Before further processing, all the cEEG recordings were decimated to have a sampling frequency of 200 Hz, and filtered with a high pass filter with cut off frequency at 0.3 Hz and a notch filter with stopband at 48 to 52 Hz. The montage used while computing all the EEG parameters was longitudinal bipolar montage consisting of 16 derivations (Fp1-F7, F7-T3, T3-T5, T5-O1, Fp1-F3, F3-C3, C3-P3, P3-O1, Fp2-F4, F4-C4, C4-P4, P4-O2, Fp2-F8, F8-T4, T4-T6, and T6-O2).

Quantitative EEG parameters

The cEEG measurements performed in this study produced multiple time series of QEEG parameters per patient. In order to compare these time series to the outcome they were compressed to a set of summary statistics for each patient. These patient-wise statistics were then compared with outcome, measured by the dichotomized GOS.

We did this by first computing a set of QEEG parameters from 1-minute long periods. Based on the parameters obtained from the 1-minute periods, we then computed QEEG parameters that summarized the EEG activity for 6-hour long periods. The parameters computed from these 6-hour periods were updated once per hour.

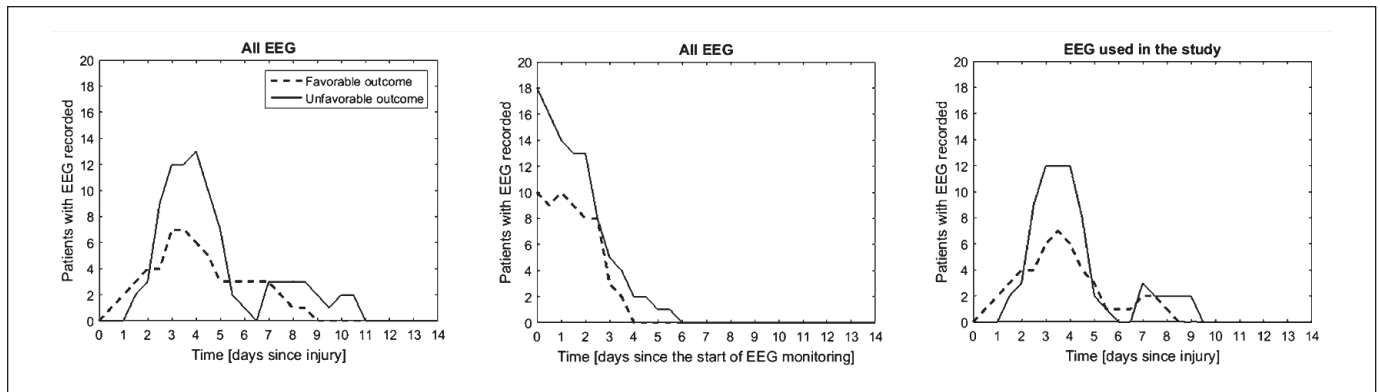


Figure 1. Distributions of the number of patients with the ongoing EEG monitoring at different time points (left: all recorded EEG using the injury time as the zero time; middle: all recorded EEG using the start of EEG monitoring as the zero time; right: the EEG used in the statistical analysis in this study; solid line: patients with unfavorable outcome; dashed line: patients with favorable outcome).

One-Minute QEEG Parameters. We computed the mean power spectral density (PSD) for each 1-minute period by first computing the PSD for consecutive 5-second epochs with 4-second overlap, and then averaging the PSDs from those epochs that contained no artifacts. From these averaged PSDs a set of parameters described below, was computed.

For automatic detection of artifactual EEG waveforms we used two criteria. First criterion examined the absolute values of EEG voltages in consecutive 10-second-long windows; if the absolute value was at any time higher than a threshold (250 μ V), EEG within that 10-second window was considered to be artifactual. This criterion detects e.g. epochs containing movement artifacts. Second criterion compared in consecutive 1-second-long windows, the signal power in frequency band 35 to 100 Hz (high frequency [HF] band) to the power in the frequency band 2 to 15 Hz (low frequency [LF] band), in order to detect periods containing HF artifacts. Based on our experience, HF artifacts were mostly caused by excessive EMG activation. If the power in HF band was higher than 10 μ V², and ratio of power in HF band to power in LF band was higher than 0.25 the EEG waveform was considered to be artifactual.

The power spectrum of the EEG was divided into following frequency bands: delta (1-3.5 Hz), theta (3.5-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), slow theta (3.5-5 Hz), fast theta (5-8 Hz), slow beta (13-20 Hz), and fast beta (20-30 Hz). For each of the bands the following spectral parameters were computed: total power in the band (parameters named: “[band name] power,” where [band name] is “delta,” “theta,” “alpha,” “beta,” “slow theta,” “fast theta,” “slow beta,” or “fast beta”), relative band power (parameters named: “relative [band name] power”), that is, power in the band divided by the total power in frequency band 1 to 30 Hz (“total power”). The PSD was also characterized by computing the mean dominant frequency, median frequency, 95% spectral edge frequency, and the spectral entropy.¹⁸ All these were computed using only the frequency band 1 to 30 Hz.

The interhemispheric asymmetry of the power spectrum was quantified by comparing the corresponding derivations

from left and right hemisphere to each other (eg, Fp1-F7 to Fp2-F8). The parameters used for this comparison were the differences in logarithms of absolute band powers (parameters named: “[band name] asymmetry” and “total asymmetry”) and differences in relative band powers (parameters named: “relative [band name] asymmetry”) for each frequency band.

To complete our analysis to cover also time-domain parameters, we computed 2 variations of the wavelet subband entropy (WSE). WSE is a QEEG parameter computed from the discrete wavelet transform of the EEG, and there is evidence that low values of WSE are connected to epileptiform activity.¹⁹ As with the PSD variables the EEG within the 1-minute period was divided into 5-second-long epochs with an overlap of 4 seconds. WSE from frequency band 16 to 32 Hz using Daubechies 3 wavelet (WSE 16-32 Hz Db3), and the total WSE²⁰ using Daubechies 1 wavelet (total WSE Db1) were computed for each 5-second epoch. The values computed from these 5-second epochs were then averaged for each derivation excluding the 5-second epochs that contained artifacts.

For each 1-minute-long period also the burst-suppression ratio (BSR)²¹ and the revised brain symmetry index (r-BSI)²² was computed. Contrary to interhemispheric asymmetry measures described above, the r-BSI is an aggregate measure that combines all symmetric comparison pairs into a single measure and it was computed using only the frequency band $1 \text{ Hz} \leq f < 30 \text{ Hz}$. Artifact criterion integrated into the BSR algorithm was used in connection to BSR values, whereas r-BSI utilized the same criterion as other PSD-based parameters. Overall, 42 QEEG parameters were derived from each 1-minute period.

6-Hour QEEG Parameters. As a measure of central tendency of QEEG parameters from the 1-minute periods, we computed the median (Med) of each QEEG parameter for the 6-hour-long periods for each of the parameters.

We also quantified EEG variability within the 6-hour epochs by computing measures of variability for the relative and absolute band powers. For the relative band power parameters, we quantified the variability by computing the ratio of median

Table 1. Grouping of the QEEG Parameters.

Parameter Group	Parameters
Absolute power	delta power, theta power, alpha power, beta power, slow theta power, fast theta power, slow beta power, fast beta power, total power
Relative power	relative delta power, relative theta power, relative alpha power, relative beta power, relative slow theta power, relative fast theta power, relative slow beta power, relative fast beta power
Asymmetry	delta asymmetry, theta asymmetry, alpha asymmetry, beta asymmetry, slow theta asymmetry, fast theta asymmetry, slow beta asymmetry, fast beta asymmetry, total asymmetry, relative delta asymmetry, relative theta asymmetry, relative alpha asymmetry, relative beta asymmetry, relative slow theta asymmetry, relative fast theta asymmetry, relative slow beta asymmetry, relative fast beta asymmetry, and revised brain symmetry index (r-BSI)
Variability	delta variability, theta variability, alpha variability, beta variability, slow theta variability, fast theta variability, slow beta variability, fast beta variability, total variability, relative delta variability, relative theta variability, relative alpha variability, relative beta variability, relative slow theta variability, relative fast theta variability, relative slow beta variability, relative fast beta variability
Other	mean dominant frequency, median frequency, 95% spectral edge frequency, spectral entropy, total wavelet subband entropy Daubechies I (WSE Db1), WSE 16-32 Hz Db3, and burst-suppression ratio (BSR)

absolute deviation (MAD) to the median. The variability of the relative alpha power computed this way is an automated method to derive an index similar to the percentage alpha variability (PAV), which is shown to be a predictor of outcome in TBI.⁸ The variability of the absolute powers was quantified by computing the MAD of the logarithms of the powers. MAD was used instead of the ratio MAD/Med, since the logarithms of the powers are not bound to be positive.

Both these variability statistics are invariant to scaling, that is, $MAD\{X\}/Med\{x\} = MAD\{aX\}/Med\{aX\}$ and $MAD\{\log(X)\} = MAD\{\log(aX)\}$. This property was deemed desirable since then the variability measures provide information that is more independent of the overall amplitude of the parameter values.

Adding the parameters measuring variability of the band powers, increased the total number of QEEG parameters computed from each 6-hour epoch to 59.

For each of the medians and the variability parameters we then computed the median over all the derivations. For the r-BSI this step was not done since the computation of r-BSI already produces only one value that contains the information from all the derivations.

Patient-wise Summary Statistics. We summarized the parameters computed from the 6-hour-long periods for each patient by computing the mean of the parameter values. The mean was computed from the 6-hour periods within first 60 hours of EEG monitoring. Periods that contained less than 3 hours of EEG were ignored.

To ease the interpretation of the results patient-wise summary statistics were grouped into 5 groups listed in Table 1.

Statistical Analysis

We analyzed patient data for possible differences between the outcome groups. For age and the medication dosages during the ICU stay, we first tested if they were normally distributed

using Anderson-Darling test; and if they were 2-sided 2-sample *t* test was used, if not 2-sided Mann-Whitney *U* test was used. For GCS at admission and Marshall CT classification Mann-Whitney *U* test was used. Fisher's exact test was used for sex, injury mechanism, and whether the patient had surgery due to an intracranial injury.

For each QEEG parameter we computed the area under the receiver operating characteristic curve (AUC), considering the favorable outcome as the positive class. In this setup, AUC is equal to the probability that a patient with favorable outcome will have a higher parameter value than patient with unfavorable outcome.²³ We also computed the *P* value for rejecting the null hypothesis that the AUC is equal to .5 using the 2-sided Mann-Whitney *U* test and the standard error for AUC using bootstrapping.

From earlier studies, age is known to be good predictor of outcome in severe TBI.²⁴ To make sure the QEEG parameters were not only measuring age related effects, we also computed the AUC for each QEEG after removing the correlation to age using linear regression.

The statistical analyses, excluding the Fisher's exact tests, were carried out using MATLAB (MATLAB R2014b, MathWorks Inc). For Fisher's exact test the implementation in R stats package version 3.1.1 was used.

The limit of statistical significance was set at the .05 *P*-value threshold.

Visual Analysis of QEEG Parameter Trends and EEG Tracings

We were interested in 2 aspects of the temporal behavior of the QEEG parameters. (1) How does the length of EEG data used in computing the patient-wise summary statistics affect the separability of the two outcome groups? (2) Does the QEEG parameter behave consistently over the recovery process; that is, if the statistical analysis shows that high values of QEEG parameters are associated with good outcome, does the good outcome group have higher parameter values in all time periods after the injury?

Table 2. Patient Characteristics in Unfavorable Outcome (Glasgow Outcome Scale, GOS 1-3) and Favorable Outcome Groups (Glasgow Outcome Scale, GOS 4-5).^a

	Unfavorable (GOS 1-3)	Favorable (GOS 4-5)	P
n	18	10	—
Age (years, mean $\pm$ SD)	52.5 $\pm$ 17.6	36.9 $\pm$ 19.9	.08
Sex: (male/female)	15/3	7/3	.63
GCS at admission (mean $\pm$ SD)	8.4 $\pm$ 4.8	9.8 $\pm$ 3.9	.41
Marshall CT classification			.18
I	0	0	
II	3	1	
III	2	0	
IV	13	8	
V	0	1	
GOS at follow-up			—
1	6	0	
2	0	0	
3	12	0	
4	0	6	
5	0	4	
Injury mechanism			.32
Road traffic	9	4	
Incidental fall	7	3	
Violence/assault	2	1	
Other/not known	0	2	
Surgery (yes/no)	7/11	6/4	.43
Medication dosages (mean $\pm$ SD)			
Propofol (mg/kg/h)	1.74 $\pm$ 1.28	1.45 $\pm$ 0.63	.49
Midazolam (mg/kg/h)	0.03 $\pm$ 0.04	0.01 $\pm$ 0.02	.68
Lorazepam (mg/kg/d)	0.02 $\pm$ 0.04	0.02 $\pm$ 0.04	.45
Fosphenytoin (mg/kg/d)	3.43 $\pm$ 2.87	2.72 $\pm$ 3.02	.56
Levetiracetam (mg/kg/d)	8.73 $\pm$ 10.73	7.02 $\pm$ 15.63	.28

^aThe *P* values of age, Glasgow Coma Scale (GCS) at admission, and Marshall CT classification are obtained from Mann-Whitney *U* test, the *P* values of injury mechanism and surgery are from Fisher's exact test, and the *P* values for the medication dosages were obtained either from 2-sided *t* test (propofol) or from Mann-Whitney *U*-test (midazolam, lorazepam, fosphenytoin, and levetiracetam).

As one cannot devise simple statistical tests to answer these questions, we performed retrospective visual analysis of temporal behavior of the best QEEG parameters found in the statistical analysis. We created 2 plots for each QEEG parameter. The first one presented the development of the AUC and the mean parameter values in the 2 outcome groups, when the amount of EEG used for computing the patient-wise mean increases. The second one presented the AUC and mean parameter value in the two outcome groups for each 6-hour period after the injury.

As a part of the routine care, a certified neurophysiologist reviewed all EEG data of this study. In addition to this primary EEG review, the neurophysiologist retrospectively reviewed EEG from patients, which had high or low values in QEEG parameters with good prognostic power found in statistical analysis. The aim was to search possible common and specific EEG patterns within groups.

Results

The demographic data of the patients showed no statistically significant differences between the outcome groups (Table 2).

All the QEEG parameters, except the parameters measuring EEG asymmetry were computed for all the subjects in the study and compared with the outcome. Four of the 28 patients had, due to the nature of their injury, an EEG montage that consisted only of electrodes on one hemisphere; and computation of the QEEG parameters assessing asymmetry between hemispheres was impossible. Of these 24 patients 8 had a favorable and 16 had an unfavorable outcome.

For 17 of the 59 (28.8%) QEEG parameters the AUC without age correction differed from 0.5 with *P* value significance level .05; these parameters are listed in Table 3.

Most of these parameters measure either absolute EEG band power or the variability of the EEG. The entire list of QEEG parameters and the associated statistics can be found in the electronic supplementary material.

All the parameters measuring absolute band power of the EEG had positive correlation to the outcome, median alpha power having the strongest correlation (AUC = 0.87, *P* < .01). The time evolution of median alpha power is illustrated in Figure 2. As more EEG data is used for estimating the parameter value there is a modest increase in the AUC values (Figure

Table 3. AUCs and the Corresponding *P* Values for all QEEG Parameters for Which AUC Without Age Correction Differs From 0.5 With *P* Value Significance Level .05.^a

Parameter Name	Parameter Group	AUC	<i>P</i>	AUC – a.c.	<i>P</i> – a.c.
Alpha power	Absolute power	0.87 ± 0.07	<.01	0.75 ± 0.10	.03
Relative fast theta power variability	Variability	0.84 ± 0.08	<.00	0.80 ± 0.09	.01
Slow beta power	Absolute power	0.84 ± 0.09	<.00	0.73 ± 0.11	.05
Total power	Absolute power	0.82 ± 0.08	.01	0.76 ± 0.09	.03
Relative alpha power variability	Variability	0.82 ± 0.09	.01	0.77 ± 0.11	.02
Fast theta power	Absolute power	0.81 ± 0.08	.01	0.72 ± 0.10	.06
Beta power	Absolute power	0.81 ± 0.08	.01	0.74 ± 0.11	.04
Delta power	Absolute power	0.81 ± 0.10	.01	0.72 ± 0.10	.06
Relative theta power variability	Variability	0.81 ± 0.08	.01	0.72 ± 0.11	.06
Alpha power variability	Variability	0.80 ± 0.09	.01	0.75 ± 0.09	.03
Slow theta power	Absolute power	0.80 ± 0.09	.01	0.68 ± 0.10	.12
Theta power	Absolute power	0.80 ± 0.09	.01	0.69 ± 0.10	.11
Fast beta power	Absolute power	0.78 ± 0.12	.02	0.73 ± 0.11	.05
Relative theta power	Relative power	0.22 ± 0.09	.02	0.28 ± 0.10	.06
Relative fast theta power	Relative power	0.23 ± 0.09	.02	0.27 ± 0.10	.05
BSR	Other	0.24 ± 0.09	.03	0.32 ± 0.10	.12
Relative slow theta power	Relative power	0.25 ± 0.11	.03	0.43 ± 0.12	.55

Abbreviations: a.c., age correction; AUC, area under receiver operating characteristic curve; BSR, burst-suppression ratio.

2A), and the patients in the favorable outcome group have higher values of median alpha power throughout the whole recovery process (Figure 2B).

Also, all parameters measuring variability had a positive correlation to the outcome. Of the variability measures the relative fast theta power variability (AUC = 0.84, *P* < .01) and the relative alpha power variability (AUC = 0.82, *P* < .01) were among the best predictors of outcome. Both the parameters showed the same desirable behavior over time as the alpha power. The time evolution of relative alpha power variability is illustrated in Figure 3.

Neurophysiologist retrospectively reviewed EEG data of 3 patients with high values and 4 patients with low values in alpha power and relative alpha power variability. Fast activity was a common and specific observation in patients with high values, that is, it was reported with all 3 patients and not mentioned with any patient with low values. EEG activity of patients with low values consisted, for example, of monotonous delta and theta activity and burst-suppression, but these findings were not specific as they appeared also in patients with high values.

Discussion

Main Findings of the Study

In this study, we presented the potential of QEEG as a prognostic tool for patients with severe TBI. Interestingly, EEG absolute power has positive correlation to the outcome, as the patients who recovered had stronger amplitude in their EEG

than those who did not. The BSR was negatively associated with outcome. Absolute power and BSR may measure the same phenomena, since the lengthening of suppression periods in EEG lowers the mean PSD. In patients with TBI, EEG can be suppressed because of sedation, or because of the trauma itself.²⁵ As our long-term goal is to develop simple and robust prognostic QEEG parameters to bedside monitoring and as we believed (and showed) that BSR would contain prognostic value, we did not exclude burst suppression periods as Vespa and coworkers⁸ did and this may have some effect to the results. Also, we did not exclude any periods based on the sedation level, as was done by Vespa et al.,⁸ as we believed that because of its impracticality it will not happen in the real clinical practice either. There was no statistically significant difference in the given drug dosages between the 2 outcome groups (Table 2). Future studies may reveal whether the inclusion or exclusion of excessive sedation periods would affect the results.

In addition, the EEG variability measures correlated with outcome; higher variability was associated with better outcome. This is in line with earlier findings,^{7,8,9} particularly the variability of relative alpha power is very similar to the percentage alpha variability described by Vespa et al.⁸ We acknowledge the merits of alpha variability measures introduced by Vespa and coworkers: For example, loss of alpha variability is a sign of alpha coma, which, especially when occurring in connection with loss of reactivity, has been showed to be an indicator of poor outcome in patients with coma.²⁶ However, contrary to the percentage alpha variability no visual rating of variability by human expert is needed in order to compute the QEEG parameters used in our study.

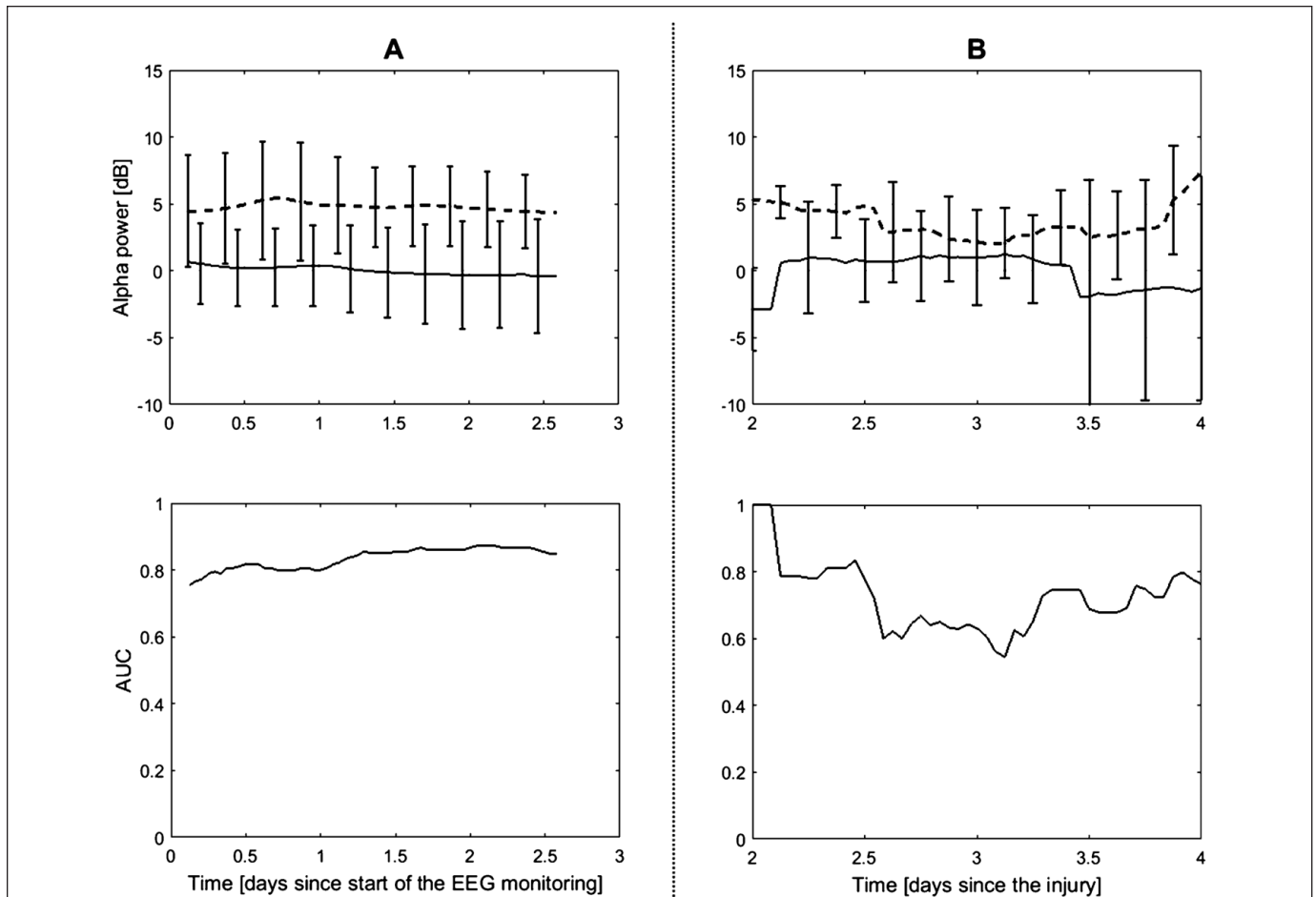


Figure 2. Illustration of the time evolution of the alpha power. (A) Mean and standard deviations for both of the outcome groups (top) and AUC (bottom) as a function of the amount of EEG available for estimating the parameter. At each time point all EEG up to that time are used for estimating the parameter values. (B) Mean and standard deviations for both of the outcome groups (top) and AUC (bottom) at different phases of the recovery process. At each time point, the preceding 6 hours are used for estimating the parameter values. In the top figures, favorable outcome group is drawn using dashed line and unfavorable outcome group using solid line. AUC, area under receiver operating characteristic curve.

The QEEG parameters measuring asymmetry seemed to have no clear connection to outcome in our study, contrary to an earlier study that demonstrated prognostic accuracy of about 80% with asymmetry and coherence measures.¹³ There are multiple possible causes for asymmetry in patients with TBI. The injury can naturally affect the electrical activity of the brain halves differently; but subcutaneous edema, hematomas and skin wounds can also affect conductivity and thickness of the tissue between the brain and the scalp electrodes.

Addressing Practical Obstacles for EEG Monitoring in ICUs

The work required for electrode attachment and monitoring of contact quality limit the use of EEG monitoring in ICUs. Abbreviated EEG montages, such as subhairline montage, are user-friendly approaches that often provide clinically meaningful information of cortical activity, although sensitivity in

seizure detection is compromised.²⁷ Future work may reveal if similar prognostic power can be obtained with abbreviated EEG montages. In this study we also used modern subcutaneous electrodes for which electrode attachment was straightforward and the contact remained stable over the monitoring period, as suggested also in an earlier study.²⁸

Limitations of the Study

This study has some limitations. The study population is still rather small, consisting only of 28 patients. The number of patients is considerably small for the open-ended type of analysis we have performed, but since very little is known about the usefulness of any QEEG parameters for the prognosis of TBI patients we still feel that this type of analysis is useful for generating more specific hypotheses for future studies. In addition, since all the EEG recordings are from a single ICU, for example by the treatment protocols used at this specific site can bias the results.

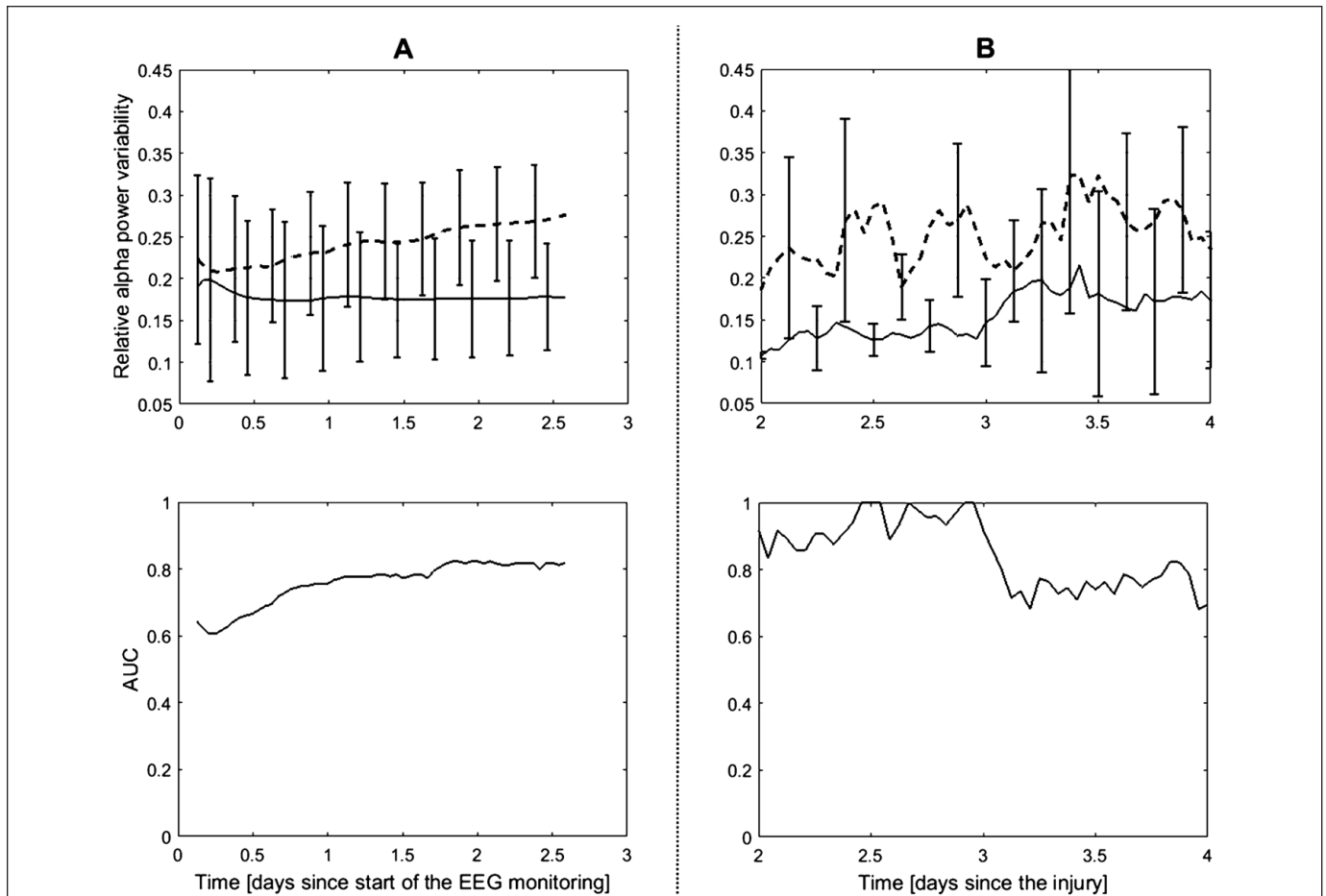


Figure 3. Illustration of the time evolution of the relative alpha power variability. (A) Mean and standard deviations for both of the outcome groups (top) and AUC (bottom) as a function of the amount of EEG available for estimating the parameter. At each time point all EEG up to that time are used for estimating the parameter values. (B) Mean and standard deviations for both of the outcome groups (top) and AUC (bottom) at different phases of the recovery process. At each time point the preceding 6 hours are used for estimating the parameter values. In the top figures, favorable outcome group is drawn using dashed line and unfavorable outcome group using solid line. AUC, area under receiver operating characteristic curve.

It is possible that confounding factors, such as age or differences in medication, explain part of the differences in QEEG parameters between the 2 outcome groups. The best QEEG parameters had a statistically significant correlation to the outcome also after removing the correlation to the age using linear regression (see Table 3), which suggests that QEEG parameters provide useful information for outcome prediction independent of the patient age. In future studies, the predictive power of the QEEG parameters should be compared with clinical predictive models such as the ones developed in IMPACT study²⁹ in order to find out the added value provided by them.

It is possible that the artifact rejection criteria used in this study have missed some artifacts; but manual removal of artifacts would have been too laborious, considering the long duration of the EEG recordings analyzed in this study. To minimize the effect that these missed artifacts have on the results, we have used statistical methods that are highly robust to outliers.

The use of median values over the derivations means that focal pathologies might go “unnoticed” by the QEEG parameters used in this study. The choice to limit our analysis only parameters that are well suited for characterizing global changes in the EEG was made out of necessity. The number of patients in our study is very limited and adding more parameters would have increased the likelihood of false positive findings in the statistical analysis.

Conclusions

To conclude, we conducted a development study for a large set of QEEG parameters against the outcome of severe TBI patients. A set of the best performing QEEG parameters should be evaluated in a different, preferably multi-center, study. Encouragingly, results of the study indicated general usefulness of EEG monitoring, as confirmed in earlier studies.⁷⁻¹³

Author Contributions

AT contributed to analysis and interpretation; drafted manuscript; critically revised manuscript; gave final approval; agrees to be accountable for all aspects of work ensuring integrity and accuracy. MS contributed to conception and design; contributed to analysis and interpretation; drafted manuscript; critically revised manuscript; gave final approval; agrees to be accountable for all aspects of work ensuring integrity and accuracy. RT contributed to conception and design; contributed to interpretation; critically revised manuscript; gave final approval; agrees to be accountable for all aspects of work ensuring integrity and accuracy. AK contributed to conception and design; contributed to interpretation; critically revised manuscript; gave final approval; agrees to be accountable for all aspects of work ensuring integrity and accuracy. JF contributed to conception and design; critically revised manuscript; gave final approval; agrees to be accountable for all aspects of work ensuring integrity and accuracy. JP contributed to interpretation; critically revised manuscript; gave final approval; agrees to be accountable for all aspects of work ensuring integrity and accuracy. MM contributed to design; contributed to interpretation; critically revised manuscript; gave final approval; agrees to be accountable for all aspects of work ensuring integrity and accuracy. MV contributed to design; contributed to interpretation; critically revised manuscript; gave final approval; agrees to be accountable for all aspects of work ensuring integrity and accuracy. OT contributed to conception and design; contributed to interpretation; critically revised manuscript; gave final approval; agrees to be accountable for all aspects of work ensuring integrity and accuracy.

Declaration of Conflicting Interests

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Supplementary Material

Supplementary Material is available for this article online.

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